GROUP WEEKLY REPORTS

Project Title: Why 2024 Flights Ran Late
Team: Team 10 – DATA-VIZ HACK-A-BIT
Team Members: Annalakshmi R, Ashvitha I, Dharshini P, Joshna Amirthaa Mariappan

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WEEK 4 – DATA CLEANING AND PREPROCESSING

During the fourth week, our team worked on preparing the dataset for further analysis by performing data cleaning and preprocessing operations. The primary objective was to improve data consistency and reliability while preserving meaningful information.

We addressed the issues identified during the previous week's data quality assessment. The team examined missing values, invalid entries, duplicate observations, and inconsistencies in numerical and categorical variables.

Using Pandas, we applied suitable methods to handle missing information and remove unnecessary duplicate records where appropriate. We also checked the data types of different columns

and ensured that the variables were represented in appropriate formats.

Special attention was given to departure delay and arrival delay values. We examined unusually large delay values and investigated whether they represented genuine flight delays or errors. We decided that legitimate extreme values should be retained because they reflect actual flight conditions and may be important for understanding delay patterns.

The team also discussed the importance of maintaining data integrity during preprocessing. We understood that careless removal or modification of observations could affect the accuracy of statistical summaries and lead to incorrect conclusions.

We reviewed the cleaned data to ensure that the preprocessing operations had been performed appropriately. The team also documented the cleaning process and the decisions made during the preparation of the dataset.

By the end of the week, we had developed a cleaner and more consistent dataset that could be used for exploratory data analysis, statistical investigation, and visualization.